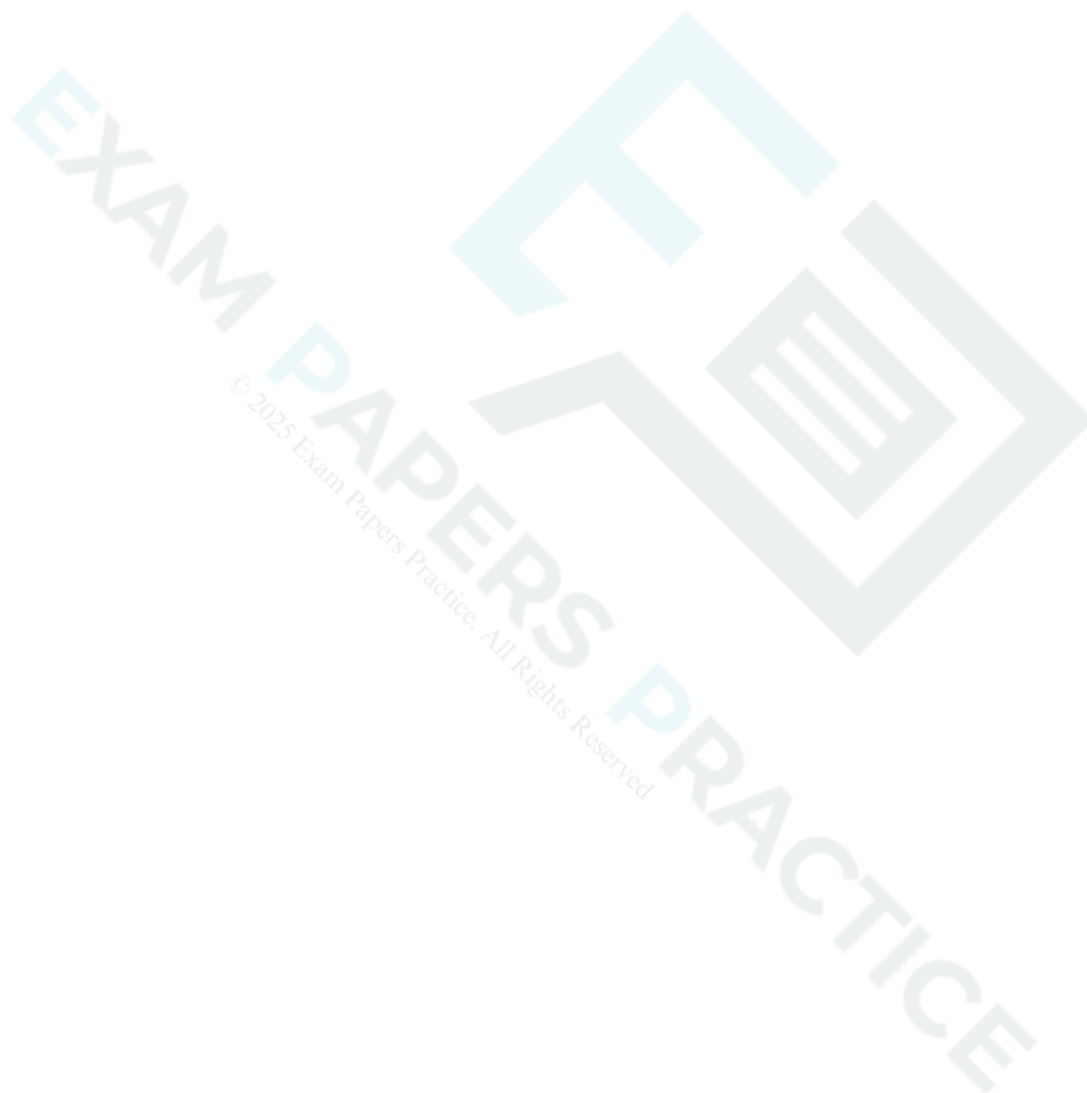
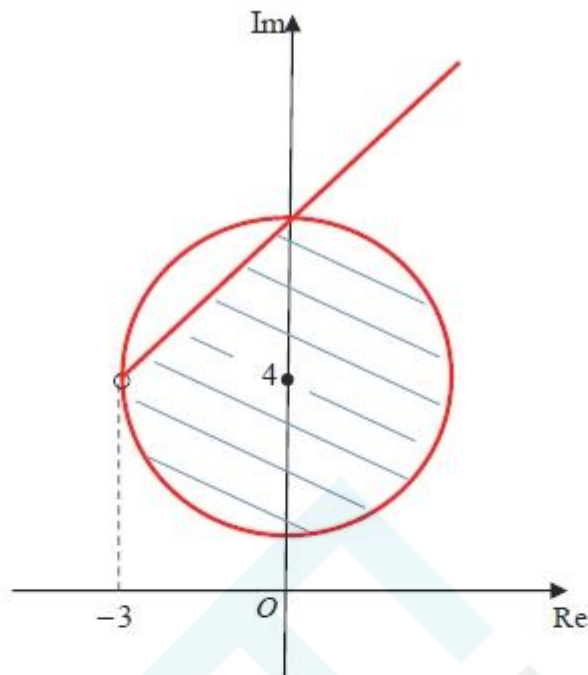




Given: Due:	Further Maths (Core Pure) HW 03	Name:
Chapter 2: Complex Numbers: loci		
(a) Shade on an Argand diagram the set of points $\left\{ z \in \mathbb{C} : z - 4i \leq 3 \right\} \cap \left\{ z \in \mathbb{C} : -\frac{\pi}{2} < \arg(z + 3 - 4i) \leq \frac{\pi}{4} \right\}$ The complex number w satisfies $ w - 4i = 3$ (b) Find the maximum value of $\arg w$ in the interval $(-\pi, \pi]$. Give your answer in radians correct to 2 decimal places.		
		(6)
		(2)





Question	Scheme		Marks	AOs
(a)			M1	1.1b
			A1	1.1b
			M1	1.1b
			A1	2.2a
			M1	3.1a
			A1	1.1b
			(6)	
(b)	$(\arg w)_{\max} = \frac{\pi}{2} + \arcsin\left(\frac{3}{4}\right)$		M1	3.1a
	$= 2.42 \text{ (2dp) cao}$		A1	1.1b
			(2)	
(8 marks)				
Question Notes				
(a)	M1	Circle.		
	A1	Centre (0, 4) and above the real axis.		
	M1	Half-line.		
	A1	(-3, 4) positioned correctly and the half-line intersects the top of the circle on the y-axis.		
	M1	Depends on both previous M marks Shades in a region inside the circle and below the half line.		
(b)	A1	Cso		
	Note	Final A1 mark is dependent on all previous marks being scored in part (a).		
	M1	Uses trigonometry to give an expression for an angle in the range $\left(\frac{\pi}{2}, \pi\right)$ or $(90^\circ, 180^\circ)$.		
	A1	2.42 cao		

(a) Shade on an Argand diagram the set of points

$$\{z \in \mathbb{C} : |z - 1 - i| \leq 3\} \cap \left\{z \in \mathbb{C} : \frac{\pi}{4} \leq \arg(z - 2) \leq \frac{3\pi}{4}\right\} \quad (5)$$

The complex number w satisfies

$$|w - 1 - i| = 3 \text{ and } \arg(w - 2) = \frac{\pi}{4}$$

(b) Find, in simplest form, the exact value of $|w|^2$

(4)

The problem solving aspect of this question proved a challenge for many candidates, with very few scoring highly on this question. There was a general confusion of what to do between parts (b) and (c), with many not seeing that the calculation via evaluation of the quotient was necessary to prove the result in part (c).

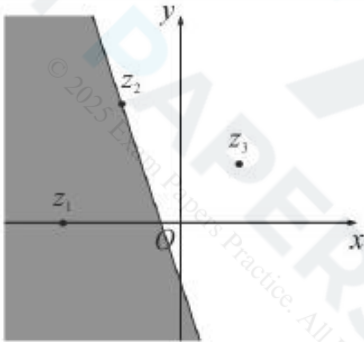
(a) The majority of candidates mentioned complex conjugates, however, very few gave a complete argument. Many failed to mention that a quartic has a maximum of 4 roots or to show that the given values would result in 5 roots.

(b) The main issue with parts (b) and (c) was candidates confusing the methods. For those who understood what was needed for part (b) the majority correctly worked through multiplying the denominator by the conjugate (or used calculator) to get to $\frac{1}{2} + \frac{1}{2}i$ and achieve the first two marks. Many lost the final A mark due to lack of justification for $\arctan(1)$ being $\pi/4$ rather than $-3\pi/4$. Those who did justify largely did so with the use of a diagram.

There were many candidates who attempted to apply the difference of arguments in this part, and subsequently used decimal approximations to try and prove the result. Such attempts scored no marks for (b) but often recovered one or both marks in part (c). There were also a few esoteric approaches, using vectors or triangle geometry, to try and establish the result.

(c) Many did not link this to the given complex numbers and the work already done in part (b), and those who did use the correct formula were often repeating work from part (b). Where decimals had been used in (b), they could recover both marks in this part as long as decimals were not referred to again, but for many, use of decimals was the default. Overall this part was not well attempted.

(d) Most candidates scored at least 1 mark here though some did not appreciate that the bisector passed through point z_2 . The majority did shade the correct side. Common errors included lines passing through zero and candidates drawing circles. Many attempted this part even if they had become stuck earlier on in the question.

Question	Scheme	Marks	AOs	
(a)	Complex roots of a real polynomial occur in conjugate pairs	M1	1.2	
	so a polynomial with z_1 , z_2 and z_3 as roots also needs z_2^* and z_3^* as roots, so 5 roots in total, but a quartic has at most 4 roots, so no quartic can have z_1 , z_2 and z_3 as roots.	A1	2.4	
		(2)		
(b)	$\frac{z_2 - z_1}{z_3 - z_1} = \frac{-1+2i-(-2)}{1+i-(-2)} = \frac{1+2i}{3+i} \times \frac{3-i}{3-i} = \dots$	M1	1.1b	
	$= \frac{3-i+6i+2}{9+1} = \frac{5+5i}{10} = \frac{1}{2} + \frac{1}{2}i$ oe	A1	1.1b	
	As $\frac{1}{2} + \frac{1}{2}i$ is in the first quadrant (may be shown by diagram), hence $\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \arctan\left(\frac{\frac{1}{2}}{\frac{1}{2}}\right) (= \arctan(1)) = \frac{\pi}{4}^*$	A1*	2.1	
		(3)		
(c)	$\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \arg(z_2 - z_1) - \arg(z_3 - z_1) = \arg(1+2i) - \arg(3+i)$	M1	1.1b	
	Hence $\arctan(2) - \arctan\left(\frac{1}{3}\right) = \frac{\pi}{4}^*$	A1*	2.1	
		(2)		
(d)		Line passing through z_2 and the negative imaginary axis drawn.	B1	1.1b
		Area below and left of their line shaded, where the line must have negative gradient passing through negative imaginary axis but need not pass through z_2	B1	1.1b
	Unless otherwise indicated by the student mark Diagram 1(if used) if there are multiple attempts.			
		(2)		
(9 marks)				

(a)	M1	Some evidence that complex roots occur as conjugate pairs shown, e.g. stated as in scheme, or e.g. identifying if $-1+2i$ is a root then so is $-1-2i$. Mere mention of complex conjugates is sufficient for this mark.
	A1	A complete argument, referencing that a quartic has at most 4 roots, but would need at least 5 for all of z_1, z_2 and z_3 as roots. There should be a clear statement about the number of roots of a quartic (e.g. a quartic has four roots), and that this is not enough for the two conjugate pairs and real root.
(b)	M1	Substitutes the numbers in expression and attempts multiplication of numerator and denominator by the conjugate of their denominator or uses calculator to find the quotient. (May be implied.) NB Applying the difference of arguments and using decimals is M0 here.
	A1	Obtains $\frac{1}{2} + \frac{1}{2}i$. (May be from calculator.) Accepted equivalent Cartesian forms.
	A1*	Uses arctan on their quotient and makes reference to first quadrant or draws diagram to show they are in the first quadrant. to justify the argument.
(c)	M1	Applies the formula for the argument of a difference of complex numbers and substitutes values (may go directly to arctans if the arguments have already been established). If used in (b) it must be seen or referred to in (c) for this mark to be awarded. Allow for $\arg(z_2 - z_1) - \arg(z_3 - z_1)$ if $z_2 - z_1$ and $z_3 - z_1$ have been clearly identified in earlier work.
	A1*	Completes the proof clearly by identifying the required arguments and using the result of (b). Use of decimal approximations is A0.
(d)	B1	Draws a line through z_2 and passing through negative imaginary axis.
	B1	Correct side of bisector shaded. Allow this mark if the line does not pass through z_2 . But it should be an attempt at the perpendicular bisector of the other two points – so have negative gradient and pass through the negative real axis.
		Ignore any other lines drawn for these two marks.

$$|z-1-i|=4, z \in \mathbb{C}.$$

a) Sketch the locus of the points that satisfy the above equation in a standard Argand diagram.

b) Find the minimum and maximum values of $|z|$ for points that lie on this locus.

(6)

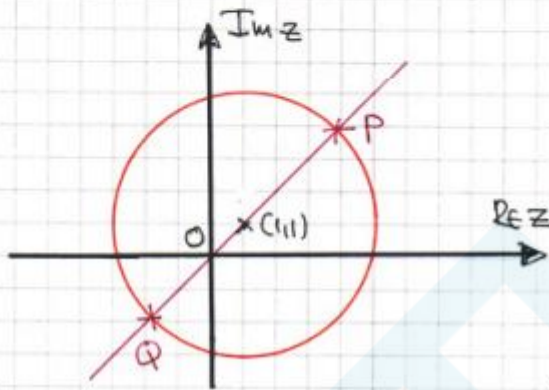
a)

THIS IS A STANDARD CIRCLE LOCUS

$$|z - 1 - i| = 4$$

$$|z - (1 + i)| = 4$$

IT CIRCLE AT $(1, 1)$, RADIUS 4



b)

LOOKING AT THE DIAGRAM ABOVE & NOTING THAT $|z|$ REPRESENTS THE DISTANCE OF A POINT FROM THE ORIGIN

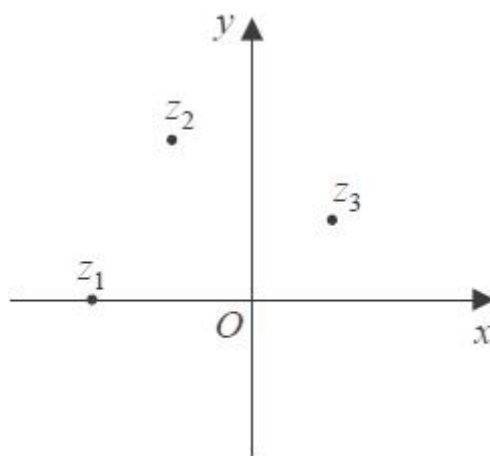
- DISTANCE OF THE CENTRE FROM O IS $\sqrt{2}$

- $|z|_{\text{MAX}} = \text{RADIUS} + \sqrt{2} = 4 + \sqrt{2}$

(POINT P)

- $|z|_{\text{MIN}} = \text{RADIUS} - \sqrt{2} = 4 - \sqrt{2}$

(POINT Q)

**Figure 1**

The complex numbers $z_1 = -2$, $z_2 = -1 + 2i$ and $z_3 = 1 + i$ are plotted in Figure 1, on an Argand diagram for the complex plane with $z = x + iy$

(a) Explain why z_1 , z_2 and z_3 cannot all be roots of a quartic polynomial equation with real coefficients.

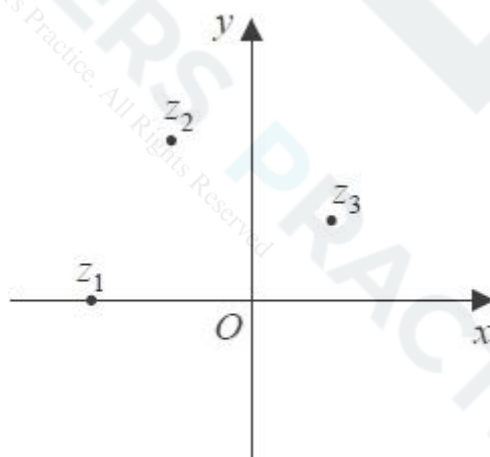
(2)

(b) Show that $\arg \left(\frac{z_2 - z_1}{z_3 - z_1} \right) = \frac{\pi}{4}$

(3)

(c) Hence show that $\arctan(2) - \arctan\left(\frac{1}{3}\right) = \frac{\pi}{4}$

(2)

**Diagram 1**

(d) Shade, on Diagram 1, the set of points of the complex plane that satisfy the inequality

$$|z + 2| \leq |z - 1 - i|$$

(2)

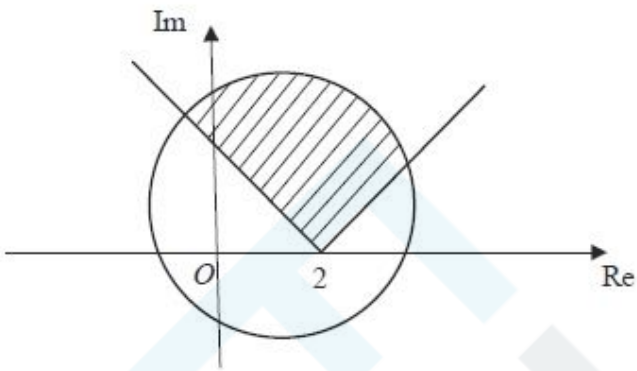
(a) The majority of students were able to produce a meaningful diagram with nearly all managing a circle with centre $1 + i$

The vertex of the "V" shape varied in position, some had it at the origin and also at various other positions. For those who did have a "V" shape, the directions of the branches also varied, with some pointing both to the left or right. Most students realised that the left hand branch crossed the y-axis below the circle although many had the left branch and the circle intersecting at the same point on the y-axis.

Irrespective of success with the circle and "V" shape, students almost always knew that they had to shade

the area between the "V" and within the circle and so could at least recover a mark following an incorrect diagram.

(b) Proved much more challenging and many students did not know where to start. Those that were able to give the equation of the circle and equation of a half line gained the method marks by attempting to solve simultaneous equations and using the solutions correctly. Some students tried to work with the diagram and find the point of intersection by simple trigonometry but often made false assumptions when doing so. A few considered the triangle with vertices at (2, 0), the centre of the circle and the point of intersection, correctly calculating the sides as $\sqrt{2}$, 3 and $\sqrt{7}$. They were then able to get the point of intersection from trigonometry. A few students left their answer as a decimal rather than a surd.

Question	Scheme	Marks	AOs
(a)		M1	1.1b
		M1	1.1b
		A1	2.2a
		M1	3.1a
		A1	1.1b
		(5)	
(b)	$(x-1)^2 + (y-1)^2 = 9, y = x - 2 \Rightarrow x = \dots, \text{ or } y = \dots$	M1	3.1a
	$x = 2 + \frac{\sqrt{14}}{2}, y = \frac{\sqrt{14}}{2}$	A1	1.1b
	$ w ^2 = \left(2 + \frac{\sqrt{14}}{2}\right)^2 + \left(\frac{\sqrt{14}}{2}\right)^2$	M1	1.1b
	$= 11 + 2\sqrt{14}$	A1	1.1b
		(4)	
(9 marks)			



(a)

M1: Circle or arc of a circle with centre in first quadrant and with the circle in all 4 quadrants or arc of circle in quadrants 1 and 2

M1: A "V" shape i.e. with both branches above the x -axis and with the vertex on the positive real axis. Ignore any branches below the x -axis.

A1: Two half lines that meet on the positive real axis where the right branch intersects the circle or arc of a circle in the first quadrant and the left branch intersects the circle or arc of a circle in the second quadrant but not on the y -axis.

M1: Shades the region between the half-lines and within the circle

A1: Cso. A fully correct diagram including 2 marked (or implied by ticks) at the vertex on the real axis with the correct region shaded and all the previous marks scored.

(b)

M1: Identifies a suitable strategy for finding the x or y coordinate of the point of intersection.

Look for an attempt to solve equations of the form $(x \pm 1)^2 + (y \pm 1)^2 = 9$ or 3 and $y = \pm x \pm 2$

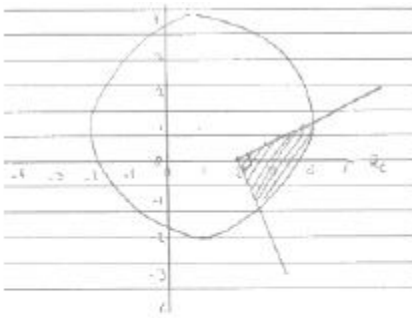
A1: Correct coordinates for the intersection (there may be other points but allow this mark if the correct coordinates are seen). (The correct coordinates may be implied by subsequent work.)

Allow equivalent exact forms and allow as a complex number e.g. $2 + \frac{\sqrt{14}}{2} + \frac{\sqrt{14}}{2}i$

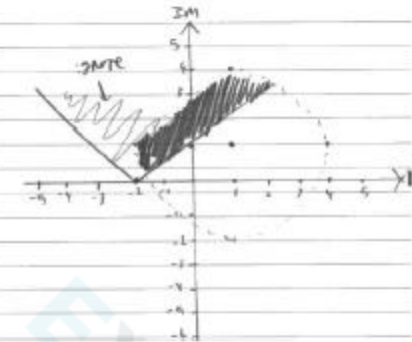
M1: Correct use of Pythagoras on their coordinates (There must be no i 's)

A1: Correct exact value by cso

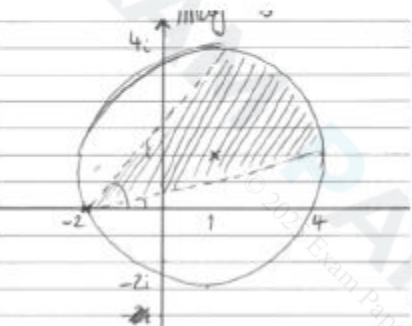
Note that solving $(x-1)^2 + (y-1)^2 = 9$, $y = x + 2$ gives $x = \frac{\sqrt{14}}{2}$, $y = 2 + \frac{\sqrt{14}}{2}$ and hence the correct answer fortuitously so scores M1A0M1A0



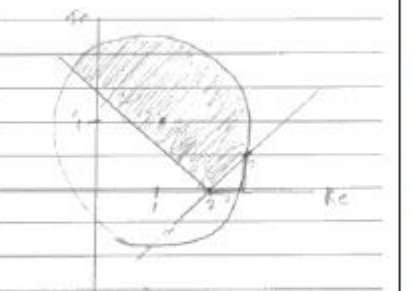
M1: Circle with centre in first quadrant
M0: The branches of the "V" must be above the x-axis
A0: Follows M0
M1: Shades the region between the half-lines and within the circle
A0: Depends on all previous marks



M1: Circle with centre in first quadrant
M0: The vertex of the "V" must be on the positive x-axis
A0: Follows M0
M1: Shades the region between the half-lines and within the circle (BOD)
A0: Depends on all previous marks



M1: Circle with centre in first quadrant
M0: The vertex of the "V" must be on the positive x-axis
A0: Follows M0
M1: Shades the region between the half-lines and within the circle
A0: Depends on all previous marks



M1: Circle with centre in first quadrant
M1: A "V" shape i.e. with both branches above the x-axis and with the vertex on the positive real axis. Ignore any branches below the x-axis.
A1: Two half lines that meet on the positive real axis where the right branch intersects the circle in the first quadrant and the left branch intersects the circle in the second quadrant.
M1: Shades the region between the half-lines and within the circle
A1: A fully correct diagram including 2 marked at the vertex on the real axis with the correct region shaded and all the previous marks scored.